

Gabrielle Pang

Singapore | gab.prq@gmail.com | +65 8797 7819 | gabyybaraa.github.io/gab-portfolio | github.com/gabyybaraa

EDUCATION

NANYANG POLYTECHNIC

Diploma in Robotics and Mechatronics, School of Engineering

2023 – 2026

Singapore

HOLY INNOCENTS' HIGH SCHOOL

GCE O-Level

2018 – 2022

Singapore

EXPERIENCE

ROBOT BARISTA ARM AUTOMATION – Overseas Industrial Training Programme

2025

Team Project | Robotic Systems & Automation

Korea

- Applied GPIO-triggered control logic between Raspberry Pi and a robotic arm to execute precise and repeatable automated drink-making sequences.
- Built and reviewed RoboDK simulations to validate motion paths, positioning and sequence flow before physical testing.
- Programmed, tested and refined predefined robotic movements through repeated runs, visual inspection and technical troubleshooting.
- Separately assembled drone components, wired circuits and configured the transmitter-receiver connection during the overseas programme.

SELECTED PROJECTS

CV AGENT – Telegram AI Productivity & Visual Reasoning Assistant

2026

Personal AI Agent Project

Singapore

- Built a Telegram-based assistant combining Qwen2.5-VL visual reasoning, Groq-powered AI chat, SQLite storage, Gmail IMAP email triage and a Flask dashboard.
- Developed workflows for reminders, notes, visual note scanning, university application tracking, scheduled briefings and AI summaries.

SMART IOT PEOPLE COUNTER – Smart IR Detection System

2025

Final Year Project | Developed for use in NYP Library

Singapore

- Engineered directional IR detection using ESP8266 microcontrollers and infrared sensors to classify entry and exit movement while reducing false counts.
- Connected sensor nodes through a central WiFi hub and displayed live occupancy data through a Flutter mobile application and web dashboard.
- Integrated AWS for cloud data transmission, storage and monitoring; conducted hardware assembly, sensor testing, debugging and validation.

CUSTOM ORDERING APP FOR SMALL BUSINESS

2026

Web Application | Firebase & Business Automation

Singapore

- Designed and developed a responsive ordering application using Firebase to manage customer accounts, products, orders and stock-related workflows.
- Built product browsing, quantity selection, order summaries, PayNow instructions and WhatsApp order-export flows for desktop and mobile users.

LEADERSHIP & ACTIVITIES

GEO COUNCIL – NANYANG POLYTECHNIC

2025

Member

Singapore

- Designed visual content for student events and incorporated feedback to maintain quality and consistency.

FLOORBALL TEAM – HOLY INNOCENTS' HIGH SCHOOL

2018 – 2022

CCA Team Member

Singapore

- Represented the school in B and C Division competitions; qualified for B Division Round 2 in 2022, the first time in school history.

AWARDS / ACHIEVEMENTS

- Smart Parking System using Arduino and sensors selected for the NYP Open House showcase (2024)
- Floorball B Division – qualified for Round 2, first in school history (2022)
- ActiveSG SFA C Division Plate & Bowl – 3rd Place (2018)

SKILLS & INTERESTS

AI & Software: Python, AI Agents, Qwen2.5-VL, Groq API, Flask, SQLite, Firebase, API Integration, OpenCV, Image Processing

IoT & Automation: ESP8266, Raspberry Pi, Arduino, GPIO, IR Sensors, AWS, WiFi Communication, RoboDK, Robotic Arms, Soldering, CAD Modelling

Professional: Testing, Debugging, Technical Documentation, Analytical Thinking, Team Collaboration, Adaptability

Languages: English (Fluent), Mandarin